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Geographic isolation and climatic heterogeneity shape the genetic diversity of Blarinellini (Eulipotyphla: Soricidae) in the Hengduan Mountains

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Abstract

The Hengduan Mountains of southwestern China constitute a classic sky-island system shaped by complex topography and climatic history. We investigated the evolutionary history of the shrew tribe Blarinellini using 272 samples representing both genera and all four recognized species. Based on mitochondrial and nuclear sequence data, we reconstructed phylogenetic relationships, estimated divergence times, and examined population genetic structure and ecological niche dynamics. Our results support an early Miocene origin of Blarinellini (~18.2 million years). Population genetic analyses recovered five and four deeply divergent geographic lineages within *Blarinella quadraticauda* and *B. wardi*, respectively. Most genetic variation was partitioned among lineages, indicating strong long-term isolation. Lineage distributions closely correspond to major river systems and montane regions, suggesting that the effects of river barriers and sky-island fragmentation may have contributed to the observed patterns of diversification. Ecological niche models identified climatically stable habitats within the Hengduan Mountains across multiple glacial–interglacial cycles, whereas demographic analyses revealed recent expansion in a subset of lineages. Together, these results suggest that river barriers, sky-island fragmentation, and climatic change may have jointly contributed to diversification in Blarinellini, and suggest that evolutionary diversity within the tribe may be substantially underestimated. These findings highlight the dual role of the Hengduan Mountains as both a refuge preserving ancient lineages and a cradle generating new diversity.

Keywords

Blarinellini, ecological niche modeling, Hengduan Mountains, phylogeography, Quaternary climate fluctuations, sky islands

Introduction

The mountainous regions of southwestern China represent one of the world's major biodiversity hotspots and harbor exceptionally high levels of species richness and endemism (Myers et al. 2000; Wang et al. 2024). Within this region, the Hengduan Mountains constitute a classic sky-island system characterized by complex topogra-

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phy, strong elevational gradients, and deep river valleys formed by major drainage systems such as the Nujiang and Lancang Rivers (McCormack et al. 2009; He and Jiang 2014). The resulting mosaic of isolated montane habitats provides ideal conditions for population divergence and lineage persistence and has been recognized as an important engine of biodiversity generation in southwestern China (He 2010; Xing and Ree 2017; Li et al. 2025).

In addition to topographic complexity, the evolutionary history of the Hengduan Mountains has been strongly influenced by geological uplift and Quaternary climatic oscillations (Hewitt 2000, 2004; González-Wevar et al. 2023; Zhang et al. 2024). Repeated cycles of habitat contraction and expansion altered population connectivity through time, promoting isolation, divergence, and demographic change in many montane organisms. Numerous phylogeographic studies have demonstrated that major rivers and sky-island fragmentation can act as important drivers of diversification in southwestern China (Li et al. 2009; He and Jiang 2014; He et al. 2019). However, most studies have focused on relatively vagile vertebrates (Landis et al. 2022; Bolívar-Leguizamón et al. 2024), and the extent to which these processes have shaped diversification in low-dispersal small mammals remains insufficiently understood (Evstafiev 2021; Yuan et al. 2024; Zhu et al. 2025).

The tribe Blarinellini (Soricidae) provides an ideal model for evaluating how geographic isolation and climate change interact to shape diversification in montane small mammals. Members of the tribe are semi-fossorial shrews distributed throughout central and southwestern China and adjacent regions of Myanmar and Vietnam (He et al. 2018; Bannikova et al. 2019; Chen et al. 2023). Fossil and molecular evidence suggest that Blarinellini originated during the Miocene and has persisted through major geological and climatic transitions across East Asia (Jin et al. 2000; Qiu and Storch 2005; He et al. 2010). Their limited dispersal ability, habitat specialization, and distribution across multiple mountain systems make them particularly suitable for evaluating the effects of geographic isolation and climatic change on lineage diversification.

Recent taxonomic studies have substantially revised the classification of Blarinellini and currently recognize two genera, *Blarinella* and *Parablarinella*, comprising four species: *Blarinella quadraticauda*, *B. wardi*, *Parablarinella griselda*, and *P. latimaxillata* (He et al. 2018; Chen et al. 2023). Despite these advances, several lines of evidence suggest that evolutionary diversity within the tribe may remain underestimated. In particular, *B. quadraticauda* exhibits extensive geographic variation in morphology, deep mitochondrial divergence among populations, and remarkable chromosomal variation across its distributional range (Jiang et al. 2003; Ye et al. 2006; Chen et al. 2012; Bannikova et al. 2019; Wan et al. 2026). These observations raise the possibility that currently recognized species boundaries do not fully capture the evolutionary history of the group. Although a recent phylogeographic study of *B. quadraticauda* provided important insights into lineage structure surrounding the Sichuan Basin (Liu et al. 2025), phylogeographic patterns across the broader distribution of the tribe remain poorly understood.

To address these knowledge gaps, we combined multilocus sequence data, population genetic analyses, divergence-time estimation, and ecological niche modeling to investigate phylogeographic diversification within Blarinellini. Specifically, we asked: (1) What are the phylogenetic relationships and population genetic structure within and among species of Blarinellini? (2) To what extent have river barriers and sky-island fragmentation contributed to lineage divergence across southwestern China? (3) How have Quaternary climatic fluctuations influenced demographic history and distribution dynamics within the tribe?

Materials and Methods

Sample collection, DNA extraction, and sequencing

We examined 272 Blarinellini samples, including 127 individuals of *Blarinella quadraticauda*, 126 of *B. wardi*, 14 of *Parablarinella griselda*, and 5 of *P. latimaxillata*, representing both recognized genera and all four currently recognized species. Samples were obtained from 95 geographic localities, spanning most of the tribe's known distribution (Fig. 1). All tissues (muscle or liver) were stored at −80 °C. Voucher specimens and associated tissues are deposited in national and provincial zoological collections. Animal sampling and handling followed the guidelines of the American Society of Mammalogists (Sikes et al. 2016) and were approved by the Animal Care and Ethics Committee of Anhui Normal University (Approval No. AHNU-ET2024005).

Total genomic DNA was extracted using a DNeasy blood and tissue kit (Tiangen, China). We amplified and sequenced two mitochondrial markers (cyt *b*, 1140 bp; 16S rRNA, ~524 bp) and three nuclear loci (ApoB, ~516 bp; BRCA1, ~770 bp; RAG2, ~750 bp) by PCR, using the primers and annealing temperatures described in Chen et al. (2023). Sequence availability varied among loci because of differences in amplification success and DNA quality. A total of 271 cyt *b* sequences were obtained from the 272 examined individuals, with one *B. quadraticauda* individual (voucher number ZIN96272; sample ID BQ95) lacking cyt *b* sequence data (Table S1). In addition, we downloaded 179 gene sequences from GenBank, including two *Blarina brevicauda* and one *Cryptotis parvus* outgroup sequences (Table S1). The 654 newly generated sequences were combined with downloaded sequences and aligned in MUSCLE (Edgar 2004), followed by manual inspection and correction in MEGA 11 (Tamura et al. 2021). Sample information and GenBank accession numbers are provided in Table S1.

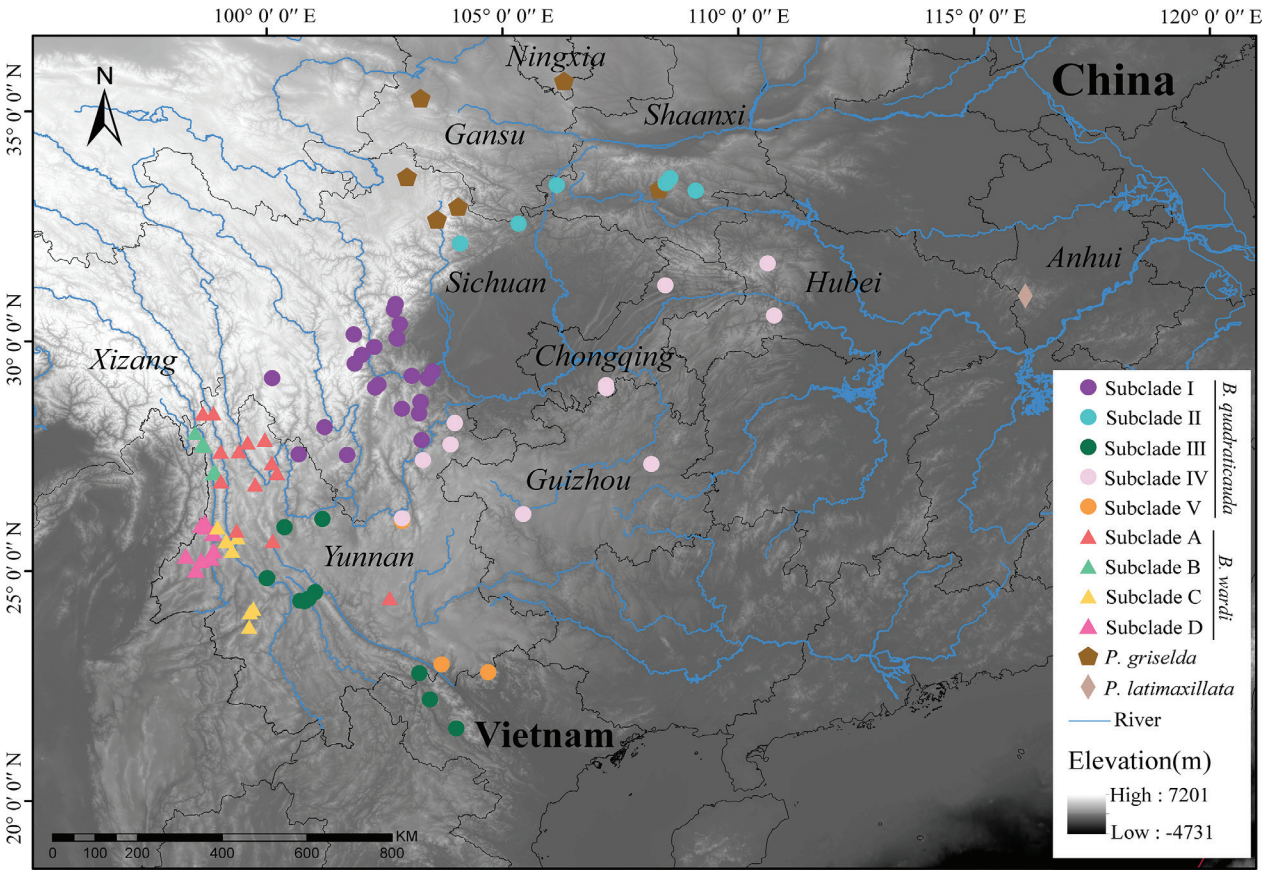


Figure 1. Sampling localities of Blarinellini included in this study. Symbols denote species, and colors indicate major subclades within *Blarinella quadraticauda* (Subclades I–V) and *B. wardi* (Subclades A–D). Major rivers are shown in blue.

Phylogenetic analyses

We inferred phylogenetic relationships within Blarinellini using Bayesian inference (**BI**) and maximum likelihood (**ML**) approaches. BI analyses were conducted in MrBayes v3.2 as implemented in PhyloSuite v1.2.2 (Zhang et al. 2020), while ML analyses were performed using RAxML via raxmlGUI 2.0 (Edler et al. 2020).

The phylogenetic reconstructions presented in the main text were based on a concatenated dataset comprising all five markers (cyt *b*, 16S rRNA, ApoB, BRCA1, and RAG2). This dataset was used to infer the primary BI and ML phylogenies and served as the basis for subsequent divergence time estimation. Optimal partitioning schemes and nucleotide substitution models were selected using PartitionFinder v2.0 (Lanfear et al. 2017) under the Bayesian Information Criterion (BIC) with a greedy search algorithm. Data blocks were defined by gene and, for protein-coding loci, by codon position, with branch lengths linked across partitions. The resulting partitioning scheme and substitution models were applied in all subsequent BI and ML analyses.

For BI analyses, two independent runs, each comprising four Markov chains, were conducted for 10 million generations, with trees and parameters sampled every 10,000 generations. Convergence was assessed using the average standard deviation of split frequencies and parameter traces, and the first 25% of samples were discarded as burn-in. ML analyses were conducted under the rapid bootstrap algorithm implemented in RAxML, with 1000 bootstrap replicates to assess node support.

To assess the consistency of phylogenetic relationships across marker types, we additionally reconstructed phylogenies from concatenated mitochondrial (cyt *b* + 16S rRNA) and nuclear (ApoB + BRCA1 + RAG2) datasets —using the same analytical framework. Single-gene trees were also inferred for each locus. These supplementary phylogenies are provided in the Figures S1–S8.

Divergence time estimation

We estimated divergence times for major nodes within Blarinellini in BEAST v2.7.7 (Bouckaert et al. 2014) using the concatenated mitochondrial and nuclear dataset. The partitioning scheme and substitution models listed in Table S2 were applied, with substitution models unlinked among partitions and a shared tree topology across all partitions. An uncorrelated lognormal relaxed molecular clock was applied to all partitions. Because the dataset included extensive intraspecific sampling and shallow population-level divergences, a coalescent constant-population prior was employed.

We implemented two fossil-based calibration points using lognormal prior distributions. The first constrained the divergence between Blarinellini and its sister tribe, Blarinini. This calibration was based on the earliest fossil occurrence of Blarinini from the Barstovian of North

America (~16.3–13.6 million years [Ma]; Repenning 1967) together with early Miocene records of Blarinellini from Europe and North America (Harris 1998; Rzebik-Kowalska 1998). The lognormal prior parameters were set to offset = 16.3, mean = 0.8, and standard deviation = 0.75, corresponding to a 95% prior probability interval of 26.0–16.8 Ma. The second calibration was based on the earliest known fossil record of *Cryptotis* (~9 Ma; Harris 1998). The lognormal prior parameters were specified as offset = 9.0, mean = 3.5, and standard deviation = 1.78, yielding a 95% prior probability interval of 16.0–9.0 Ma.

Markov chain Monte Carlo (MCMC) analyses were run for 100 million generations, sampling every 20,000 generations. We conducted two independent runs and discarded the first 10% of samples from each as burn-in. Convergence and mixing were evaluated in Tracer v1.7.2 (Rambaut et al. 2018), and all parameters achieved effective sample sizes (ESS) > 200. A maximum clade credibility (MCC) tree was generated in TreeAnnotator v2.7.7 using median node heights.

Population genetic structure and historical dynamics

Because sampling localities for *P. griselda* and *P. latimaxillata* were limited, we conducted population genetic analyses only for *B. quadraticauda* and *B. wardi*. Due to missing cyt *b* data for one *B. quadraticauda* individual, cyt *b*-based analyses included 126 individuals of *B. quadraticauda*. Analyses were performed separately for the mitochondrial cyt *b* gene and the nuclear ApoB gene, which provided greater sample coverage than the remaining nuclear loci.

For each marker, haplotypes were identified and standard diversity indices, including the number of haplotypes (H), haplotype diversity (Hd), nucleotide diversity (π), number of segregating sites (S), and average number of nucleotide differences (k), were calculated in DnaSP v5.10 (Librado and Rozas 2009). Missing data were handled using pairwise deletion, and nucleotide diversity was calculated based on the available sites within the alignment. Population genetic structure was evaluated using molecular analysis of variance (AMOVA) and pairwise F_{ST} in Arlequin v3.5.2 (Excoffier et al. 2010). Historical demographic patterns were assessed using mismatch distribution analyses in DnaSP and neutrality tests (Tajima's D and Fu's Fs; Tajima 1989; Fu 1997) in Arlequin. Statistical significance was evaluated using 1000 permutations where applicable. Missing data were treated by pairwise deletion in both DnaSP and Arlequin. Sample sizes for each subclade are provided in Table S1.

Haplotype networks for cyt *b* and ApoB were constructed using PopART v1.7 (Leigh and Bryant 2015) under the median-joining algorithm (Bandelt et al. 1999). Alignment gaps and ambiguous nucleotide sites were treated as missing data and excluded from network inference. Genetic distances (p distance) among lineages were calculated in MEGA 11 (Tamura et al. 2021).

Ecological niche modeling and distributional stability analysis

To evaluate whether the distributions of Blarinellini are consistent with long-term climatic niche stability, ecological niche models (ENMs) were constructed for *B. quadraticauda* and *B. wardi* using MaxEnt. Potential distributions were reconstructed for the Last Interglacial (LIG; 120–140 ka), the Last Glacial Maximum (LGM; ~22 ka), and the present. Occurrence records were compiled separately for each species from verified localities. To reduce spatial sampling bias and autocorrelation, duplicate records were removed and occurrences were spatially rarefied by retaining a single record within each 5 × 5 km grid cell. After filtering, 121 occurrence records for *B. quadraticauda* and 122 occurrence records for *B. wardi* were retained.

The background extent for model calibration was defined to approximate the accessible area (M) of Blarinellini, encompassing the known fossil and extant distribution range of the group, with the Yellow River as the northern boundary, the Indochinese Peninsula as the southern boundary, the Brahmaputra River as the western boundary, and the East China Sea as the eastern boundary. Nineteen bioclimatic variables were obtained from WorldClim v2.1 (Fick and Hijmans 2017) for current climatic conditions at a resolution of 2.5 arc minutes. Corresponding LGM and LIG climatic layers were obtained at the same or higher resolution and resampled to 2.5 arc minutes prior to analysis.

To reduce multicollinearity among predictors, variable selection was conducted in three steps. First, an initial MaxEnt model including all 19 variables was used to assess preliminary variable importance. Second, highly correlated variables ($|r| \geq 0.9$) were identified using Pearson correlation analysis and removed. Third, variance inflation factor (VIF) analysis was used to exclude variables with VIF values > 10. Variables with consistently low contributions in the preliminary model were subsequently excluded. The final predictor set for *B. quadraticauda* included bio2, bio4, bio10, bio13, bio14, bio18, and bio19, whereas that for *B. wardi* included bio2, bio3, bio5, bio14, bio15, bio18, and bio19.

Model complexity was optimized separately for each species by evaluating combinations of feature classes and regularization multipliers. Optimal settings were selected using the lowest Akaike Information Criterion corrected for small sample size (AICc) and subsequently used for projections under present, LGM, and LIG climatic conditions. Final models were run with 10 replicates, using 75% of occurrence records for training and 25% for testing. Model performance was evaluated using the area under the receiver operating characteristic curve (AUC). All analyses were conducted in R v4.3.3 (R Core Team 2024) with MaxEnt v3.4.3, and optimal model settings are provided in Table S3.

To assess distributional stability through time, habitat suitability predictions for the LIG, LGM, and present were converted to binary maps using a threshold of 0.5. Suitable areas (≥ 0.5) were overlaid across time periods

to identify stable and unstable habitats. Cells predicted as suitable in all three periods were classified as long-term stable habitats ("Throughout"), those suitable in two periods as relatively stable habitats, those suitable in one period as unstable habitats, and those unsuitable in all periods as "Never suitable". Following previous studies, resistance values of 1, 10, 100, and 1000 were assigned to these categories, respectively. Stability analyses were performed in ArcGIS v10.2. Variable contributions for each species are provided in Table S4.

Results

Phylogenetic relationships and divergence times

Both ML and BI analyses of the combined datasets yielded highly congruent topologies, supporting the reciprocal monophyly of *Blarinella* and *Parablarinella* (posterior probability [PP]/ bootstrap values [BS] = 1.0/100; Figs 2, S1–S3) and their respective sister-species pairs. This interspecific framework was fully recovered by individual mitochondrial (Figs S4–S5) and ApoB gene trees (Fig. S6). In contrast, single-locus nuclear markers consistently lacked intraspecific resolution, and individual RAG2 (Fig. S7) and BRCA1 (Fig. S8) loci failed to support the monophyly of *B. quadraticauda* and *P. griselda*, likely due to slower evolutionary rates.

Within *B. quadraticauda*, five geographically structured lineages (Subclades I–V) were identified. The basal lineage (Subclade V) occurred in the eastern Yunnan Plateau and northern Vietnam, whereas the remaining lineages were distributed across the Yunnan–Guizhou Plateau, the margins of the Sichuan Basin, the Qinling Mountains, and the northern Yunnan–southwestern Sichuan highlands. Within *B. wardi*, four geographically structured lineages (Subclades A–D) were recovered. Subclade A occurred east of the Lancang River, Subclade B occupied the upper Nujiang drainage west of the river, Subclade C was distributed between the Lancang and Nujiang rivers, and Subclade D occurred in the lower Nujiang drainage.

Molecular dating estimated the stem and crown ages of Blarinellini at ~18.2 Ma (95% HPD: 21.6–16.5 Ma; Fig. 3) and ~14.0 Ma (95% HPD: 18.6–9.8 Ma), respectively. Interspecific divergences were dated to ~5.3 Ma (95% HPD: 7.6–3.5 Ma) for *B. quadraticauda–B. wardi* and ~3.0 Ma (95% HPD: 4.6–1.8 Ma) for *P. latimaxillata–P. griselda*. Following initial intraspecific splits at ~2.3 Ma in both species, internal lineages diversified rapidly through the Pleistocene (Fig. 3). *Blarinella quadraticauda* exhibited a sequential divergence of Subclades II, I, and III between ~2.3 and 0.9 Ma, whereas *B. wardi* split into Subclades D, C, and the sister pair A–B from ~2.3 down to 1.0 Ma, with terminal node radiation for both taxa concluding in the mid- to late-Pleistocene. The divergence-time analysis recovered the same major subclades as the phylogenetic analyses, although the branching order of the major *B. quadraticauda* ns PD, Solovyeva EN, Pavlova SV, Demidova TB, Simanovsk

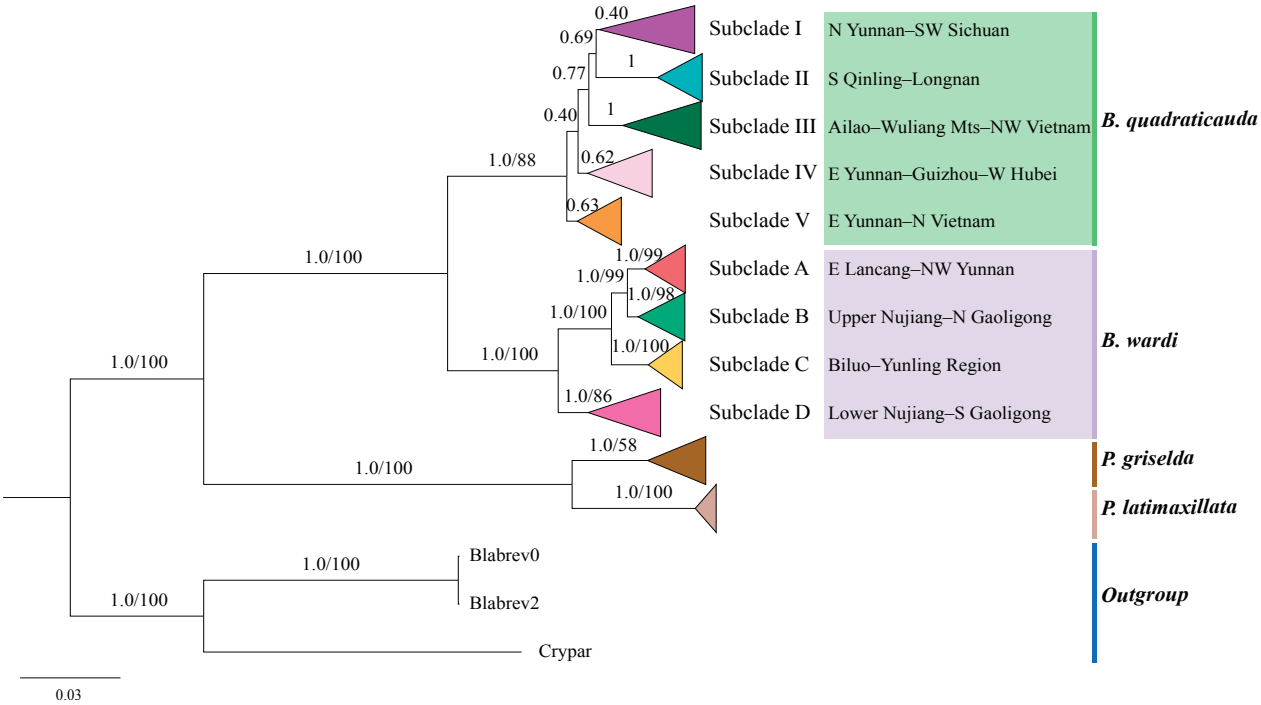


Figure 2. Bayesian inference (BI) phylogeny of Blarinellini reconstructed from the combined mitochondrial and nuclear dataset. Only major intraspecific subclades are shown. Numbers above or below branches correspond to Bayesian posterior probabilities (PP) and maximum likelihood bootstrap (BS) values, shown as PP/BS where the corresponding nodes are recovered in both analyses; for nodes not recovered in the ML tree, only PP values are shown. The scale bar denotes substitutions per site. Subclade colors correspond to Figure 1. Abbreviations: E, eastern; W, western; N, northern; S, southern; SW, southwestern; NW, northwestern.

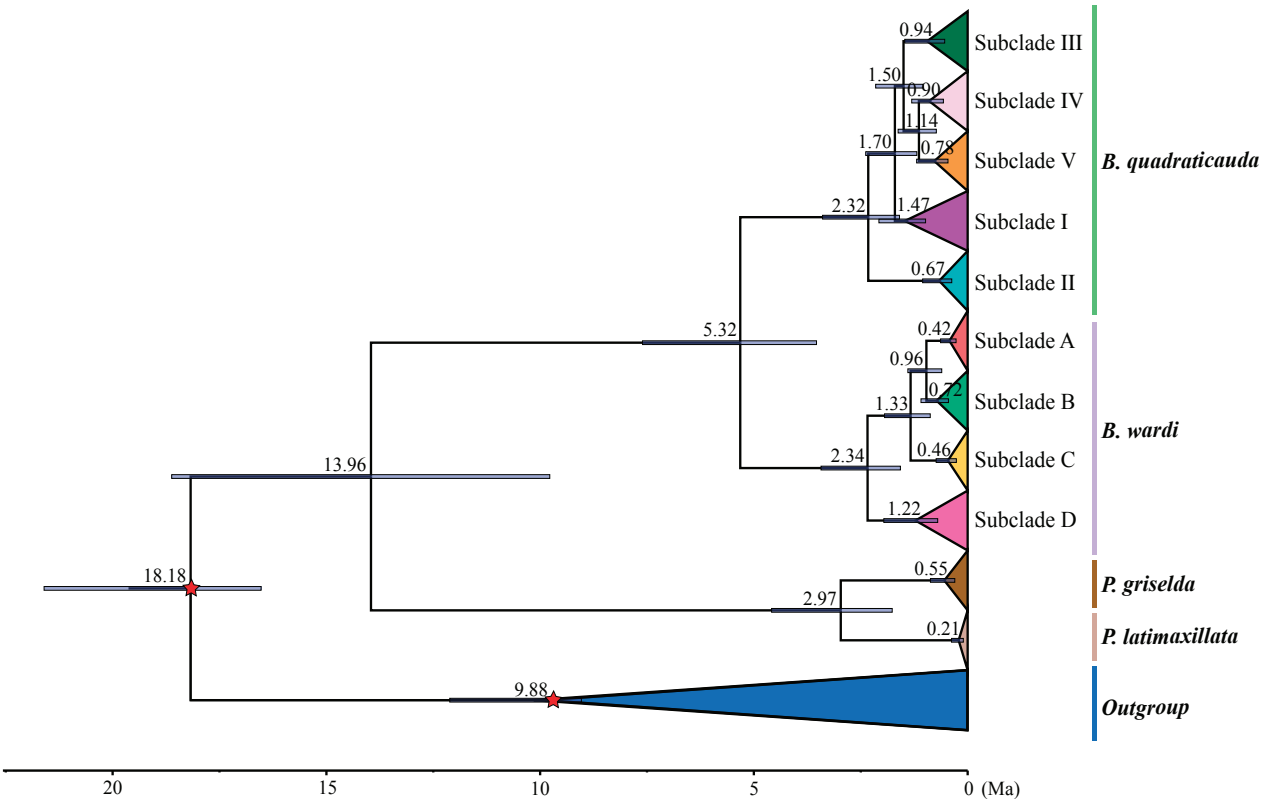


Figure 3. Divergence-time estimates within Blarinellini based on combined mitochondrial and nuclear sequences. Node labels indicate median divergence times (Ma), and bars denote 95% highest posterior density (HPD) intervals. Red stars indicate fossil-based calibration points. The time scale is in million years (Ma). Subclade colors correspond to Figure 1.

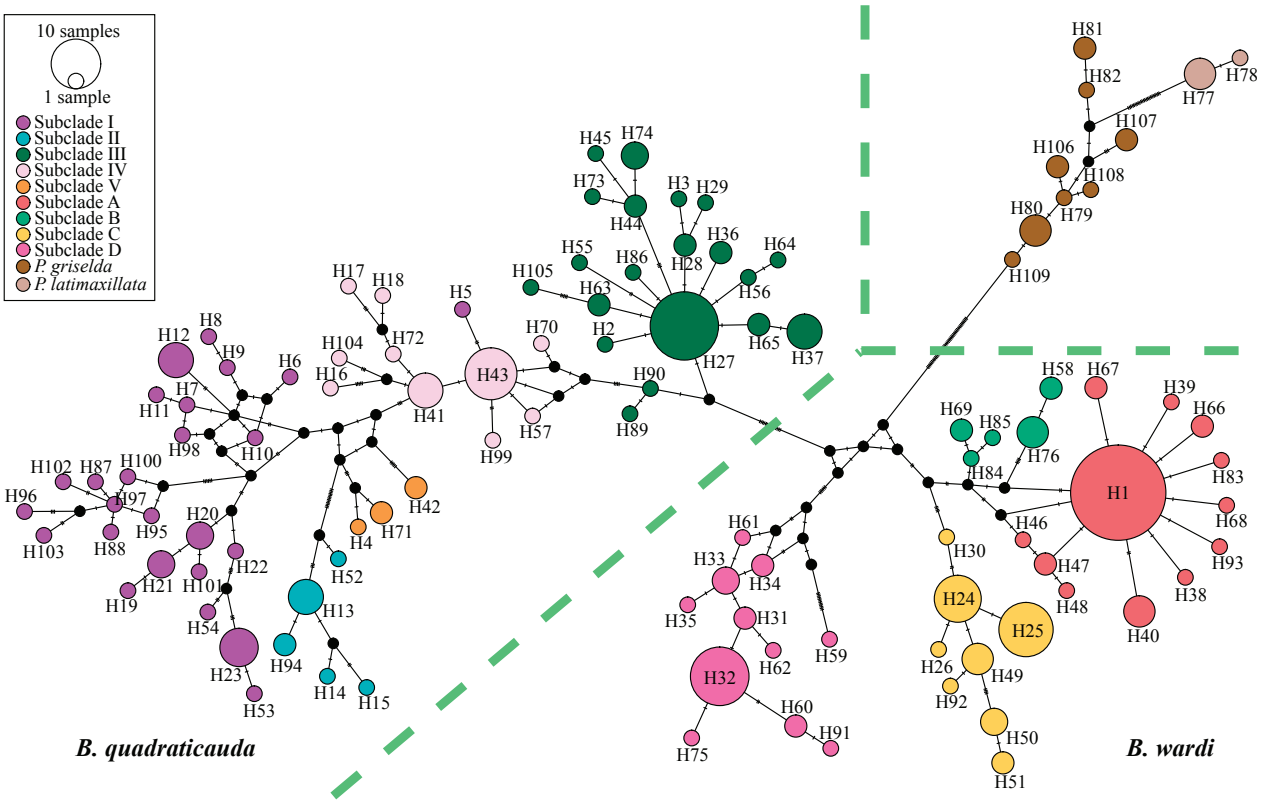


Figure 4. Median-joining network of cyt *b* haplotypes in Blarinellini. Circle size is proportional to haplotype frequency. Tick marks indicate single mutational steps. Small black circles denote inferred (unsampled) intermediate haplotypes. Colors correspond to subclades in Figures 2 and 3.

Table 1. Analysis of molecular variance (AMOVA) for populations.

Populations	Source of variation	d.f.	Sum of squares	Variance components	Percentage of variation	p value
B. quadraticauda (Subclade I–V)	Among Subclades	4	447.468	4.86848	66.77	$p < 0.001$
	Within Subclades	121	293.175	2.42293	33.23	$p < 0.001$
	Total	125	740.643	7.29141		
B. wardi (Subclade A–D)	Among Subclades	3	587.099	6.73483	87.52	$p < 0.001$
	Within Subclades	122	117.115	0.95996	12.48	$p < 0.001$
	Total	125	704.214	7.69478		

Table 2. Population genetic parameters for subclades within Blarinellini.

Species	Subclade	N	H	Hd	S	π	k	Tajima's D	Fu's F_S
B. quadraticauda	I	38	33	0.989	131	0.0298	33.186	0.0621	−6.6448*
	II	10	8	0.956	28	0.0083	9.400	−1.1614	−0.5275
	III	49	22	0.855	29	0.0080	2.895	−1.8209*	−12.2857**
	IV	24	15	0.866	72	0.0092	10.428	−1.9689**	-2.9205
	V	5	3	0.800	37	0.0184	20.800	1.7890	2.6780
B. wardi	A	54	30	0.961	51	0.0050	5.391	−2.0056**	−9.6325**
	B	10	7	0.933	34	0.0139	15.733	1.2797	1.0718
	C	33	15	0.919	34	0.0050	5.500	−0.4387	−0.5845
	D	29	21	0.963	66	0.0078	8.823	−1.6675*	−2.6780
P. griselda	—	14	11	0.967	33	0.0092	9.780	0.0420	−1.2031
P. latimaxillata	—	5	5	1.000	13	0.0049	5.600	−0.8165	0.0902

Note: N, sample size. H, number of haplotypes. Hd, haplotype diversity. S, number of polymorphic sites. π, nucleotide diversity. k, mean number of nucleotide differences. Significance of neutrality tests (Tajima's D and Fu's F_S): p* < 0.05; p** < 0.01.

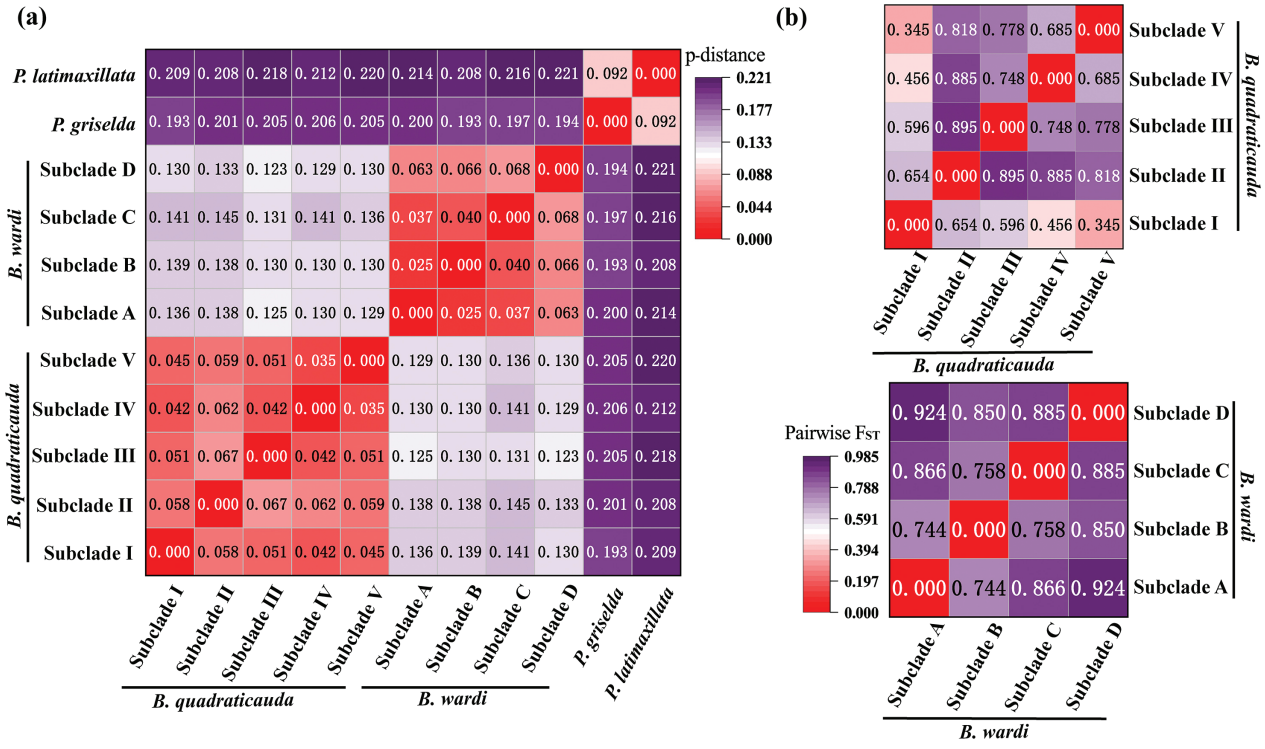


Figure 5. a Genetic distance (p distance) based on cyt *b* sequence data for Blarinellini populations. **b** Pairwise F_{ST} values among subclades of *Blarinella quadraticauda* and *B. wardi* based on cyt *b* sequence data. All pairwise F_{ST} values were significant ($p < 0.001$). Full pairwise p distance and F_{ST} matrices based on cyt *b* are provided in Table S7, and the corresponding matrices based on ApoB are provided in Table S8.

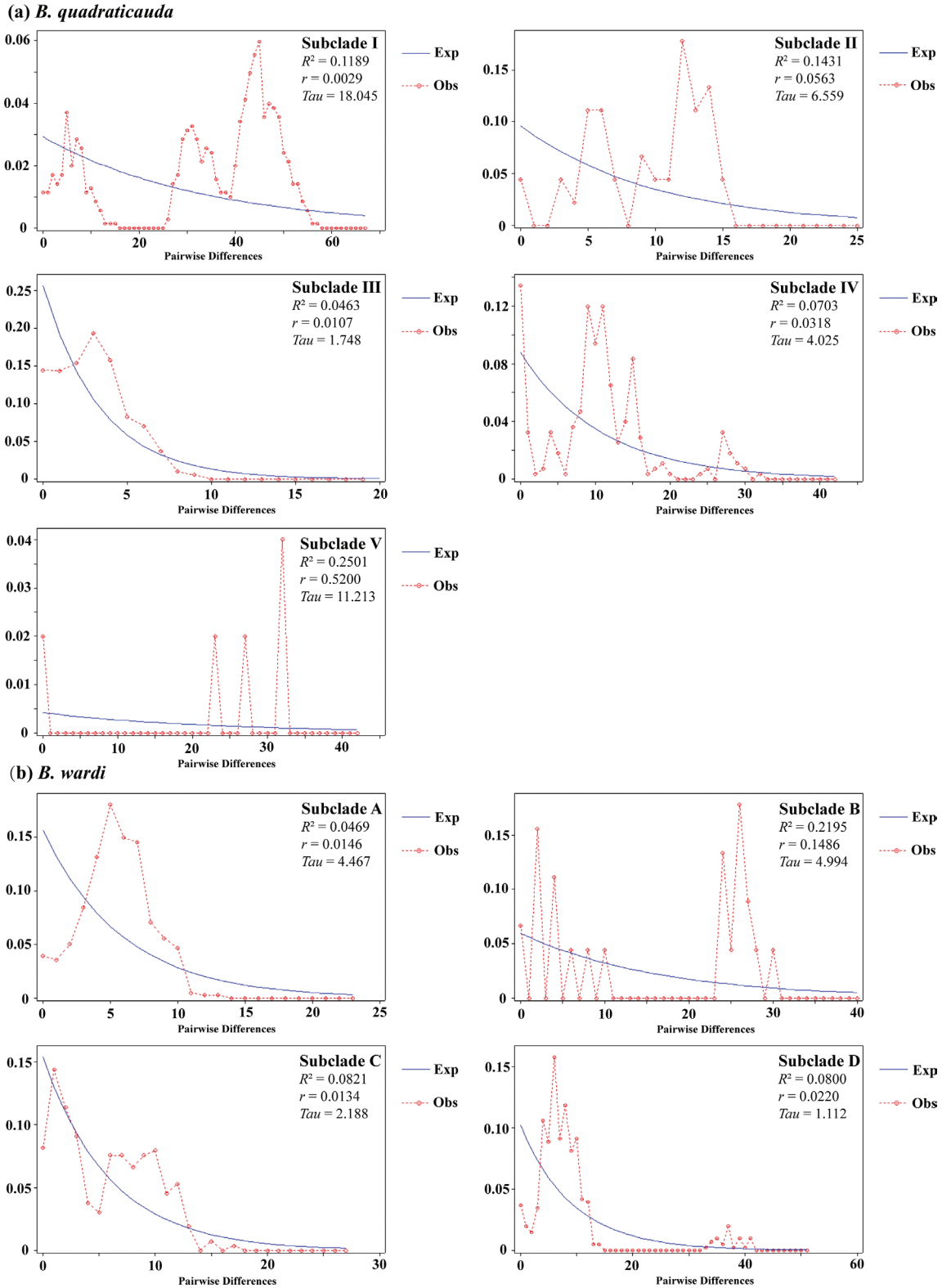


Figure 6. Mismatch distribution analyses for each subclade of *Blarinella quadraticauda* and *B. wardi* based on the cyt *b* sequences. The solid line shows the observed distribution of pairwise differences, whereas the dashed line shows the expected distribution under a sudden population expansion model.

Population genetic structure and phylogeographic patterns

Analysis of 271 cyt *b* sequences identified 109 haplotypes (H1–H109; Table S1). The median-joining network recovered four well-separated haplotype groups corresponding to *B. quadraticauda*, *B. wardi*, *P. griselda*, and *P. latimaxillata* (Fig. 4). No haplotypes were shared among species.

Within *B. quadraticauda*, five distinct mitochondrial haplogroups corresponding to Subclades I–V were recovered. These haplogroups were separated by multiple

mutational steps and numerous inferred intermediate haplotypes, indicating substantial genetic differentiation among geographic regions. Similarly, four well-defined mitochondrial haplogroups corresponding to Subclades A–D were identified within *B. wardi*. The distribution of haplotypes within both species showed strong geographic structure, with no haplotypes shared among major subclades.

Parablarinella exhibited markedly lower levels of mitochondrial variation. *Parablarinella griselda* comprised a compact cluster of closely related haplotypes, whereas *P. latimaxillata* contained only two closely related haplotypes. Despite their low internal diversity, both species remained strongly differentiated from *Blarinella*.

The nuclear ApoB dataset exhibited markedly reduced variation compared to cyt *b* and produced a shallow haplotype network (Fig. S9). Two primary haplotype groups corresponding to *B. quadraticauda* and *B. wardi* were identified, but several haplotypes were shared among mitochondrial subclades. Although overall geographic structure was weaker than in the mitochondrial dataset, a number of haplotypes were restricted to particular regions, especially within *B. quadraticauda* Subclades II and IV and *B. wardi* Subclade A.

AMOVA based on cyt *b* indicated pronounced population subdivision in both widespread species (Table 1). Most genetic variation was partitioned among subclades, accounting for 66.77% of total variation in *B. quadraticauda* (F_{ST} = 0.6677, p < 0.001) and 87.52% in *B. wardi* (F_{ST} = 0.8753, p < 0.001). Pairwise F_{ST} values and genetic distances further demonstrated substantial divergence among mitochondrial lineages (Fig. 5), whereas differentiation inferred from ApoB was considerably weaker (Table S5–S6).

Haplotype diversity was high in most subclades (Table 2). Significant negative Tajima's D and Fu's Fs, together with unimodal mismatch distributions, were observed only in *B. quadraticauda* Subclade III and *B. wardi* Subclade A (Fig. 6), indicating recent demographic expansion. Other subclades showed nonsignificant or inconsistent signals. ApoB diversity was much lower, with significant negative neutrality values limited to *B. quadraticauda* Subclades III and IV (Fig. S10; Table S5).

Ecological niche modeling and habitat stability

Ecological niche models showed high predictive performance for both species (*B. quadraticauda*: AUC = 0.919–0.956; *B. wardi*: AUC = 0.987–0.989), and the predicted present-day suitable areas were generally consistent with the observed occurrence records (Fig. 7).

The analysis predicted extensive suitable habitat for *B. quadraticauda* across southwestern and central China

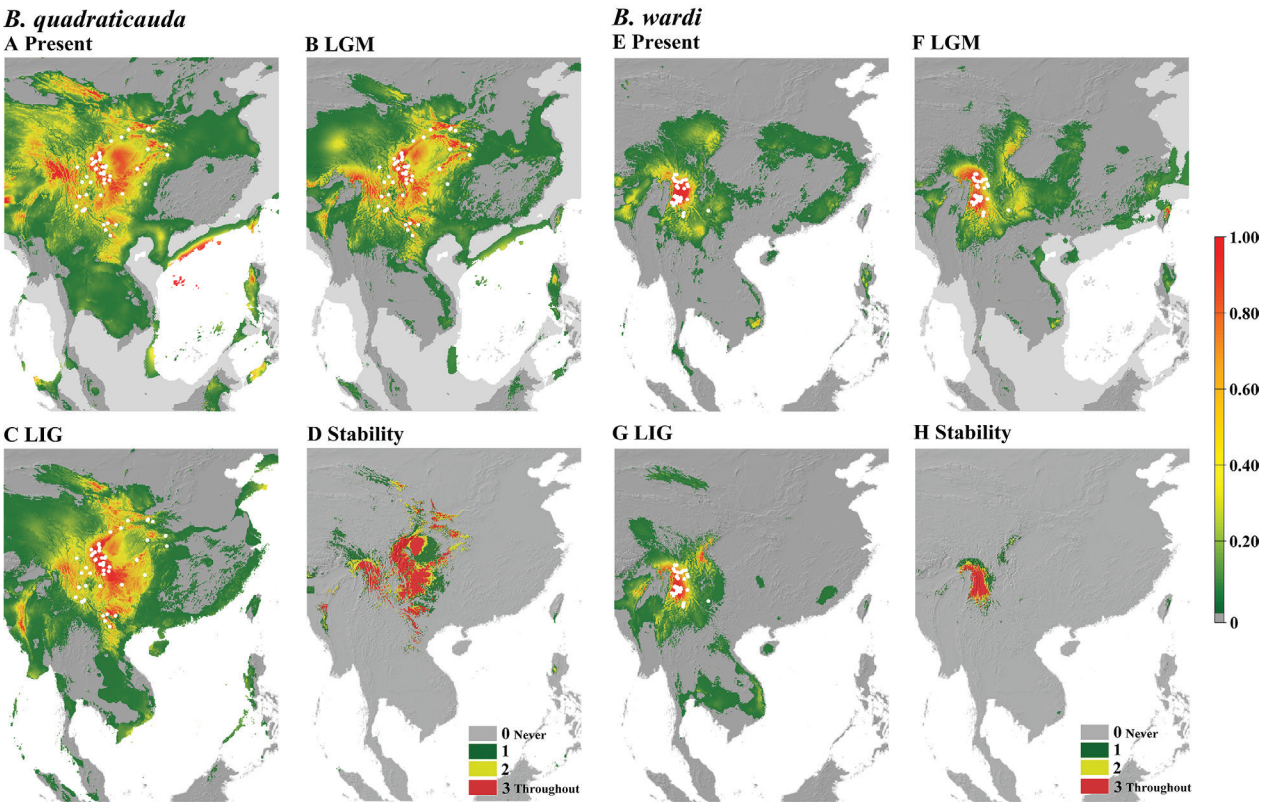


Figure 7. Ecological niche model and climatic stability analyses for *Blarinella quadraticauda* and *B. wardi*. **A**–**D** Predicted habitat suitability and climatic stability for *B. quadraticauda*: **A** present (1970–2000), **B** Last Glacial Maximum (LGM; ~22 ka), **C** Last Interglacial (LIG; ~130 ka), and **D** long-term climatic stability. **E–H** Predicted habitat suitability and climatic stability for *B. wardi*: **E** present, **F** LGM, **G** LIG, and **H** long-term climatic stability. Habitat suitability values range from 0 to 1, with warmer colors indicating higher suitability. White dots show occurrence records of Blarinellini used in this study.

under all climatic scenarios (Fig. 7A–D). Suitable habitat was most extensive during the LGM and became more fragmented during the LIG. Areas predicted to remain suitable throughout all three periods were concentrated in the Hengduan Mountains, eastern Tibetan Plateau, and adjacent montane regions, indicating long-term climatic stability across these areas.

In contrast, suitable habitat for *B. wardi* was consistently more restricted and concentrated within the southern Hengduan Mountains and southeastern Tibetan Plateau (Fig. 7E–H). Although moderate shifts in distribution were observed among climatic periods, areas of long-term stability were largely confined to the southern Hengduan Mountains. Stable habitats across all three climatic periods were identified within the Hengduan Mountains for both species.

Discussion

Our multilocus analyses agree with the currently recognized taxonomy of Blarinellini, confirming the reciprocal monophyly of the four recognized species. The phylogeographic structure within both species, however, runs far deeper than current taxonomy reflects: we recovered five mitochondrial lineages in *B. quadraticauda* and four in *B. wardi*, with no haplotypes shared among major subclades. This deep genetic structure coincides with extensive chromosomal variation. The diploid number (2n) of *B. quadraticauda* ranges from 34 to 49 (Ye et al. 2006; Bannikova et al. 2019; Wan et al. 2026), and some cytotypes correspond broadly to the mitochondrial lineages identified here. Such concordance suggests that genetic and chromosomal divergence may have accumulated in parallel during prolonged geographic isolation (Funk and Omland 2003), although whether these populations represent independently evolving lineages requires further genomic and cytogenetic evaluation. However, the extent to which chromosomal rearrangements may contribute to reproductive isolation remains unclear, as comparable karyotypic divergence in other mammals does not necessarily prevent gene flow (Faria and Navarro 2010; Carbone et al. 2014). Genomic and cytogenetic analyses will be required to evaluate the evolutionary significance of this variation (Zink and Barrowclough 2008; Ambu et al. 2023).

Our results are broadly consistent with the phylogeographic framework proposed by Liu et al. (2025). Specifically, our Subclade II corresponds to their northern lineage (Clade N), Subclade III to the southern lineage (Clade S), Subclades IV–V to the eastern lineage (Clade E), and Subclade I to the western lineages (W1–W4). The strong correspondence between these independently derived phylogeographic patterns suggests that the major lineage divisions within *B. quadraticauda* represent stable historical subdivisions rather than artifacts of sampling or analytical approach. By incorporating populations from the Hengduan Mountains, Yunnan–Guizhou Plateau, Qinling Mountains, and northern Vietnam, our study further extends this framework across a substantially larger portion of the species' distribution. In addition, the discovery of the deeply divergent Subclade V from eastern Yunnan and northern Vietnam highlights previously unrecognized evolutionary diversity outside the geographic scope of Liu et al. (2025). Together, these findings suggest that diversification in *B. quadraticauda* may have been associated with regional-scale landscape evolution across southwestern China rather than being restricted to processes within a single mountain system or geographic region.

The diversification of Blarinellini appears to have involved two major climate-associated stages (Tapponnier et al. 2001; Ding et al. 2020; Spicer et al. 2020). The first was the split between *B. quadraticauda* and *B. wardi* at approximately 5.3 Ma, around the Miocene–Pliocene boundary, which corresponds to regional cooling and aridification at that boundary (An et al. 2001; Qi et al. 2023). The second stage accounts for most intraspecific diversification: it began around 2.3 Ma and continued throughout the Pleistocene, coinciding with the development and intensification of Quaternary glacial–interglacial oscillations (Zhao et al. 2013; He et al. 2019; Li et al. 2025). Repeated fluctuations in habitat connectivity may have fragmented ancestral populations among isolated mountain systems, while subsequent range shifts could have promoted lineage accumulation within both species. The broadly synchronous histories of *B. quadraticauda* and *B. wardi* are consistent with a shared influence of regional environmental change and Quaternary climatic oscillations.

Spatially, the diversification patterns of the two species appear to have been shaped by different landscape features, with a major biogeographic boundary coinciding with the distributional separation of *B. quadraticauda* and *B. wardi*. The boundary between the two species corresponds broadly to the paleo-Jinsha River, mirroring a pattern observed in the Chinese long-tailed mole (Xing and Ree 2017; He et al. 2019). The two species otherwise respond very differently to finer-scale barriers. *Blarinella wardi* shows a pattern concordant with major river drainages, with an allopatric, highly structured distribution partitioned by major river channels: Subclade A occurs east of the Lancang River, Subclade C between the Lancang and the Nu, and Subclades B and D west of the Nu. This spatial concordance suggests that the deeply incised Lancang and Nu rivers may have contributed to limiting dispersal and gene flow (Li et al. 2009; Chen et al. 2012; He and Jiang 2014; He et al. 2019). *Blarinella quadraticauda*, by contrast, appears to show a bidirectional northward radiation from southern populations in northern Vietnam and southern Yunnan (Subclades III and V). The northeastern route, represented by Subclade IV, expanded from east of the Red River toward the Daba Mountains (Guizhou, Chongqing, and Hubei); the western route ran from Subclade III through central Yunnan and northward along western Sichuan (Subclade I) into the Qinling Mountains (Subclade II). Diversification in *B. quadraticauda* may therefore have involved parallel

colonization along separate eastern and western montane pathways, while the unsuitable lowlands of the Sichuan Basin likely restricted gene flow between the diverging lineages (Hou et al. 2024).

Evidence for recent demographic expansion was detected only in *B. quadraticauda* Subclade III and *B. wardi* Subclade A (Tajima 1989; Fu 1997), whereas most other lineages showed non-significant or inconsistent signals. Subclade III occupies the central Yunnan Plateau and the Ailao–Wuliang Mountains, whereas Subclade A is distributed mainly within the Yunling region and adjacent mountains of northwestern Yunnan. Ecological niche models indicated that suitable habitat for both species expanded during the Last Glacial Maximum and that areas of long-term climatic stability were concentrated in the Hengduan Mountains and adjacent highland regions (Hewitt 2000, 2004; He and Jiang 2014; Zhang et al. 2024). The occurrence of expansion signals in these two lineages may reflect demographic growth associated with periods of increased habitat availability in these montane regions. In contrast, the absence of comparable signals in other lineages suggests long-term persistence within more isolated mountain systems. These results indicate that demographic responses to Quaternary climatic fluctuations varied among lineages and were likely influenced by local geographic and climatic conditions rather than by a uniform range-wide response (Hewitt 2000, 2004; González-Wevar et al. 2023).

Several limitations should be acknowledged. Although our analyses revealed pronounced phylogeographic structure and deep mitochondrial divergence within both *B. quadraticauda* and *B. wardi*, inference of evolutionary history was based primarily on mitochondrial data, with limited resolution from nuclear markers, resulting in relatively low support for some deeper nodes within *B. quadraticauda* in the nuclear gene trees. Consequently, the relative contributions of incomplete lineage sorting, historical introgression, and long-term reproductive isolation cannot yet be fully resolved. Moreover, while some lineages exhibit substantial genetic divergence and correspond broadly to previously reported chromosomal variation, the taxonomic significance of these patterns remains uncertain. Future studies integrating genome-wide, morphological, and cytogenetic data will be necessary to evaluate species boundaries and clarify the evolutionary processes underlying diversification within Blarinellini.

Conclusions

Using comprehensive geographic sampling across the distribution of Blarinellini, we reveal extensive phylogeographic structure and previously unrecognized evolutionary diversity within both *B. quadraticauda* and *B. wardi*. Divergence-time analyses indicate that diversification originated during the Neogene and that major intraspecific diversification intensified during the Pleistocene, whereas population genetic patterns and ecological niche models suggest that river barriers, sky-island fragmentation, and climatic fluctuations may have jointly contributed to lineage divergence and persistence. Although the current taxonomy of two genera and four species is supported, the deep genetic structure recovered within both widespread species indicates that evolutionary diversity within Blarinellini is likely underestimated. More broadly, the concordance between phylogeographic structure, divergence history, and long-term climatic stability highlights the dual role of the Hengduan Mountains as both a refuge preserving ancient lineages and a cradle generating new diversity. These findings provide a framework for understanding how geological history, climatic change, and geographic isolation interact to generate biodiversity in mountain hotspots and highlight the potential for substantial cryptic diversity within montane small mammals.

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Supplementary Material 1

Figures S1–S10

Authors: Diao H, Xiong Y, Nie W, Pei X, Song W, Wang X, Onditi KO, Wang H, Li Q, Li X, He K, Chen Z, Jiang X (2026)

Data type: .docx

Explanation notes: Figure S1. Phylogenetic trees of Blarinellini reconstructed from the combined mitochondrial and nuclear dataset. Left: Bayesian inference (BI) tree with posterior probabilities shown at nodes. — **Figure S2.** Phylogenetic trees of Blarinellini reconstructed from the concatenated mitochondrial dataset (cyt *b* + 16S rRNA). — **Figure S3.** Phylogenetic trees of Blarinellini reconstructed from the concatenated nuclear dataset (ApoB + BRCA1 + RAG2). — **Figure S4.** Phylogenetic trees of Blarinellini reconstructed from the mitochondrial cyt *b* gene. Left: Bayesian inference (BI) tree with posterior probabilities shown at nodes. — **Figure S5.** Phylogenetic trees of Blarinellini reconstructed from the mitochondrial 16S rRNA gene. Left: Bayesian inference (BI) tree with posterior probabilities shown at nodes. — **Figure S6.** Phylogenetic trees of Blarinellini reconstructed from the nuclear ApoB gene. — **Figure S7.** Phylogenetic trees of Blarinellini reconstructed from the nuclear RAG2 gene. — **Figure S8.** Phylogenetic trees of Blarinellini reconstructed from the nuclear BRCA1 gene. — **Figure S9.** Median-joining network of ApoB haplotypes in Blarinellini. Circle size is proportional to haplotype frequency. — **Figure S10.** Mismatch distribution analyses for each subclade of *Blarinella quadraticauda* and *B. wardi* based on ApoB sequence data.

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Link: https://doi.org/10.3897/vz.76.e188776.suppl1

Supplementary Material 2

Tables S1–S8

Authors: Diao H, Xiong Y, Nie W, Pei X, Song W, Wang X, Onditi KO, Wang H, Li Q, Li X, He K, Chen Z, Jiang X (2026)

Data type: .xlsx

Explanation notes: Table S1. Sample information, voucher specimens, and GenBank accession numbers for the Blarinellini specimens used in this study. — **Table S2.** Partition schemes and substitution models selected for divergence time estimation in BEAST2. — **Table S3.** Percent contribution of the bioclimatic variables used in the final MaxEnt ecological niche models for *B. quadraticauda* and *B. wardi*. — **Table S4.** Optimal MaxEnt model settings for *B. quadraticauda* and *B. wardi*. — **Table S5.** Population genetic parameters based on ApoB sequences for subclades within Blarinellini. — **Table S6.** Analysis of molecular variance (AMOVA) based on ApoB sequences for populations. — **Table S7.** Pairwise F_{ST} (below diagonal) and genetic distance (p distance; above diagonal) based on cyt *b* sequence data for Blarinellini populations. — **Table S8.** Pairwise F_{ST} (below diagonal) and genetic distance (p distance; above diagonal) based on ApoB sequence data for Blarinellini populations.

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